Mymuna Sultana

Software Developer & Data Engineer



PERSONAL DETAILS

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Online	linkedin.com/in/mymunasultana · github.com/smymuna
Location	Fairfax, VA, USA (relocating to Germany)
Work status	Eligible for EU Blue Card (MS Computer Science) · Available for interviews in CET

PROFILE

Software Developer and Data Engineer with an M.S. in Computer Science who builds **Python** backend services, REST APIs and batch and streaming data pipelines with **FastAPI**, **Flask**, **Airflow**, **dbt** and **Kafka** on **PostgreSQL** and **AWS**. Owns work from design through **pytest** testing, **GitHub Actions** CI/CD and **Docker** deployment.

EDUCATION

01/2024 – 12/2025 Master of Science in Computer Science · George Mason University, Fairfax, VA, USA
Relevant coursework: Database Systems, Distributed Systems, Cloud Computing, Data Structures & Algorithms

2017 – 2021 Bachelor of Technology in Computer Science · Kakatiya University, Warangal, Telangana, India

PROFESSIONAL EXPERIENCE

12/2025 – present Software Developer · Mason LIFE Program, George Mason University, Fairfax, VA

- Own development of a **Python**/**Flask** and **PostgreSQL** application that centralises course, scheduling, student and program operations, replacing fragmented staff workflows with one system.
- Designed the relational data model and backend structure linking program records, courses and schedules, and consolidated **6** separate **Excel** trackers into the database to cut duplicate entry and inconsistent records.
- Build backend routes and REST endpoints for data retrieval, updates, validation and workflow execution, turning staff requirements into reusable application logic.
- Built **Python** (**pandas**), **SQL** and **dbt** workflows that clean and combine operational and student-performance data from **10+** courses into analysis-ready datasets.
- Scheduled these recurring data and reporting workflows with **Apache Airflow**, saving **4–5** hours of manual work per week.

08/2025 – 12/2025 Data Engineer Intern · DQLabs, Remote

- Engineered **Python**/**FastAPI** profiling and validation services running **50+** automated data-quality checks across **20+** datasets, using **Apache Spark** for larger datasets, reducing manual validation effort by about **30%**.
- Traced recurring production failures through structured logs, adding automatic retries that cut repeat failures by about **20%** and alerting that cut MTTR by about **35%**.
- Implemented **RabbitMQ** and **Amazon SQS** job queues to run long-running processing outside API requests, and consumed **Apache Kafka** pipeline events to trigger data-quality checks.
- Built an **AWS** data layer with **S3** for raw datasets, **PostgreSQL**/**RDS** for metadata and profiling results and **EC2** for compute, organised into raw, processed and analysis-ready layers.
- Contributed to a team-built monitoring system that profiles row counts, null rates and schema drift across **15** tables, storing metrics in **PostgreSQL** with anomaly detection and **Slack** alerts.
- Wrote **pytest** unit and integration tests, built **GitHub Actions** CI/CD workflows, and containerised **Python** services with **Docker**, supporting **Kubernetes**-based deployments.

PROJECTS

Streaming — FeedWatch · **Python**, **Apache Kafka**, **Quix Streams**, **TimescaleDB**, **FastAPI**, **React**

- Built **5** small services in one **Docker** image that stream Coinbase and Kraken trades through **Kafka**, detect feed incidents and push live updates to a **React** dashboard over Server-Sent Events.
- Streamed live Coinbase and Kraken trades through **Kafka**, with one-minute windowed metrics in **Quix Streams** and idempotent upserts into **TimescaleDB** for at-least-once delivery.
- Replayed **269,430** historical trades from the March 2023 USDC depeg as a regression test, alongside **60** unit tests and an end-to-end test through **Kafka** and **PostgreSQL**.

Batch pipeline — OrbitWatch · **Python**, **Dagster**, **dbt**, **DuckDB**, **GitHub Actions**

- Built a space-traffic pipeline that snapshots about **19,000** tracked orbits from CelesTrak every **6** hours and keeps the full history immutable in raw, **Parquet** (bronze) and **dbt** staging and mart layers.
- Modelled object history as an SCD2 slowly changing dimension built from snapshots, plus **dbt** marts for manoeuvres, re-entries, launches and owners in a **DuckDB** warehouse, orchestrated as **Dagster** assets.
- Screened **19,301** objects for close approaches with SGP4 propagation, checked against brute-force propagation; **36** tests at **93%** coverage, **dbt** tests and **Dagster** asset checks.

Full stack — GridShift · **Python**, **FastAPI**, **Next.js**, **TypeScript**, **Docker**

- Built a carbon-aware job scheduler with a **FastAPI** REST API, a CLI and a **Next.js** web app that recommends the lowest-carbon time and country for a compute job across **26** European countries.
- Wrote a window-search algorithm verified against brute-force search, and an API client that retries with exponential backoff, honours Retry-After and caches each country for **15** minutes.
- Generated a typed **TypeScript** client from the OpenAPI schema in a **Turborepo** monorepo, with CI failing on contract drift; **69 Python** tests with **90%**+ branch coverage enforced.

Backend & data — Climate Emissions Platform · **Python**, **FastAPI**, **PostgreSQL**, **React**, **Docker**

- Built a backend that ingests **3** public climate and emissions sources, normalises records and serves greenhouse-gas indicators for **32** European countries through versioned **FastAPI** REST endpoints.
- Structured **PostgreSQL** into ingest, core and mart schemas with **Alembic** migrations; shared quality rules drop, reject or flag records, and every run stores its quality report.
- Wrote **52** tests (about **92%** coverage) including integration tests on real **PostgreSQL**, with a monthly **GitHub Actions** refresh and **FastAPI** data services in **Docker**.

SKILLS & LANGUAGES

Programming	**Python**, **SQL**, **Java**, **JavaScript**, **TypeScript**, **React**, **Next.js**, **HTML**, **CSS**
Backend & APIs	**FastAPI**, **Flask**, **Spring Boot**, REST APIs, API Design, **RabbitMQ**, **Amazon SQS**
Data engineering	ETL/ELT, Data Pipelines, Data Modelling, Data Warehousing, Data Profiling, Data Quality & Validation
Pipeline tools	**Apache Airflow**, **Dagster**, **dbt**, **Apache Spark**, **Apache Kafka**, **Quix Streams**
Databases	**PostgreSQL**, **MySQL**, **MongoDB**, **DynamoDB**, **DuckDB**, **TimescaleDB**
Cloud & DevOps	**AWS** (**S3**, **RDS**, **EC2**, **Lambda**), **Docker**, **Kubernetes**, **Git**, **GitHub Actions**, CI/CD
Testing & practice	**pytest**, Unit & Integration Testing, Logging & Monitoring, OOP, Agile/Scrum
Languages	English (professional working proficiency) · German (beginner, A1)

Fairfax, October 2026

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